

N_1

A-com ring

Let $x \in A$ - nilpotent.

- ① Show that $1+x$ is invertible
- ② Deduce that the sum of a nilpotent and an invertible el. is invertible.

Proof.

x -nilpotent, then $x^n = 0$ for some $n > 0$

- ① Find $y \in A$ s.t. $(x+1)y = 1$.

$$x \cdot y + 1 \cdot y = 1$$

$$\text{Let } y = 1 - x + \dots + (-1)^{n-1} x^{n-1}$$

$$x(1 - x + \dots + (-1)^{n-1} x^{n-1}) + 1 - x + \dots + (-1)^{n-1} x^{n-1} =$$

$$= x - x^2 + \dots + (-1)^{n-1} \underbrace{x^n}_0 + 1 - x + \dots + (-1)^{n-1} x^{n-1} =$$

$$= 1.$$

Then for each nilpotent x the inverse element is $\sum_{i=0}^{n-1} (-1)^i x^i$.

② Let y is invertible. Show that $x+y$ is invertible.

$$x+y = x \cdot 1 + y = x \cdot y^{-1} \cdot y + y = (xy^{-1} + 1)y.$$

xy^{-1} is a nilpotent, because

$$(xy^{-1})^n = x^n (y^{-1})^n = 0 \cdot (y^{-1})^n = 0.$$

Then, according to ① $xy^{-1} + 1$ is invert.



$$f = a_0 + a_1 x + \dots + a_n x^n \quad f \in A[x] \quad \checkmark$$

a) Prove that f -invertible in $A[x] \iff a_0$ is invertible in A and $a_1, \dots, a_n$ -nilpot.

Proof.

$\Rightarrow f$ invertible, $g = b_0 + b_1 x + \dots + b_m x^m$ is an inverse, i.e. $f \cdot g = 1$.

Consider $f \cdot g$.

$$f \cdot g = a_0 b_0 + (a_0 b_1 + a_1 b_0)x + \dots + x^{n+m} \sum_{i+j=n+m} a_i b_j = 1$$

Then $a_0 b_0 = 1$ (i.e. a_0 is invertible)

for each $0 < k \leq n+m$ $\sum_{i+j=k} a_i b_j = 0$.

Prove that a_n is a nilpotent.

$$\text{Consider } 0 = \sum_{i+j=m+n} a_i b_j = a_n b_m$$

$$0 = \sum_{i+j=m+n-1} a_i b_j = a_{n-1} b_m + a_n b_{m-1} \quad | \cdot a_n$$

$$0 = a_n a_{n-1} b_m + a_n^2 b_{m-1}$$

$$0 = a_{n-1} \underbrace{a_n b_m}_0 + a_n^2 b_{m-1} \quad \text{Hence } a_n^2 b_{m-1} = 0$$

Now prove by induction that for all $r \geq 0$ $a_n^{r+1} b_{m-r} = 0$.

1. for $r=0$, we saw $a_n b_m = 0$

2. let for some r and for all $0 \leq k < r$ $a_n^{k+1} b_{m-k} = 0$

3. consider r

$$0 = \sum_{i+j=m+n-r} a_i b_j = a_n b_{m-r} + a_{n-1} b_{m-r+1} + \dots + a_{n-r} b_m \quad | \cdot a_n$$

$$0 = a_n^2 b_{m-r} + \dots + \underbrace{a_n a_{n-r}}_{=0} b_m \quad | \cdot a_n$$

$$0 = a_n^3 b_{m-r} + \dots + \underbrace{a_n^2 a_{n-r+1}}_{=0} b_{m-1}$$

...

$$0 = a_n^r b_{m-r}.$$

Then, by induction $a_n^r b_{m-r} = 0$ for all r .

Consider $r=m$: $a_n^m b_0 = 0 \quad | \cdot a_0$

$$a_n^m b_0 a_n = 0 \cdot a_0$$

$$a_n^m \cdot 1 = 0$$

Then a_n is a nilpotent.

Consider $f - a_n x^n = f + (-a_n x^n)$.

$(-a_n x^n)^m = (-1)^m \underbrace{a_n^m}_{=0} x^{nm} = 0$. Then

$-a_n x^n$ is a nilpotent.

Then $f + (-a_n x^n)$ is invertible, acc. to (1).

Using the same argument, a_{n-k} are nilpotents for all $0 \leq k < n$.

⇐ Let a_0 be invertible and $a_1, \dots, a_n$ nilpotents. Show that f is invertible
 $f = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$. For each $i \leq n$
 $(a_i x^i)^k = a_i^k x^{i \cdot k} = 0 \cdot x^{i \cdot k} = 0$, for such k that $a_i^k = 0$.

According to (1) a sum of inv. a_0 and nilpotents is invertible. Then f is invertible



b) f is nilpotent $\Leftrightarrow a_0, \dots, a_n$ nilpotents.

Proof.

$\Leftarrow a_0, \dots, a_n$ - nilpotents.

$$f = a_0 + c_1 x + \dots + c_n x^n.$$

$a_i x^i$ is a nilpotent. as $(a_i x^i)^k = a_i^k x^{ik} = 0$
for some k .

Lemma. a is a nilpotent of n degree
 b is a nilpotent of m degree.

Prove that $a+b$ is a nilpotent of $n+m$.

$$(a+b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^{n+m-k} b^k = \underbrace{a^n}_{=0} \underbrace{a^m}_{=0} + \underbrace{a^n}_{=0} a^{m-1} b + \dots \\ \dots + a \underbrace{b^{m-1}}_{=0} b + \underbrace{b^n}_{=0} b^m = 0.$$

Then f is a nilpotent as a sum
of a nilpotents

$\Rightarrow f$ is a nilpotent. prove $a_0, \dots, a_n$ are nilpotents.

$f^m = 0$ for some m .

$$0 = f^m = \underbrace{[\text{multinomial formula for } m-1]}_0 + \underbrace{C_n^m x^{nm}}_0.$$

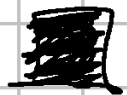
Then :

C_n^m is a nilpotent of degree m .

$-C_n^m x^{nm}$ is a nilpotent of degree m .

$f + (-C_n^m x^{nm})$ is a nilpotent. (Lemme)

using the same argument, all a_i are nilpotents,



c) f is a zero divisor $\Leftrightarrow$ for some $a \in A$ $af = 0$.

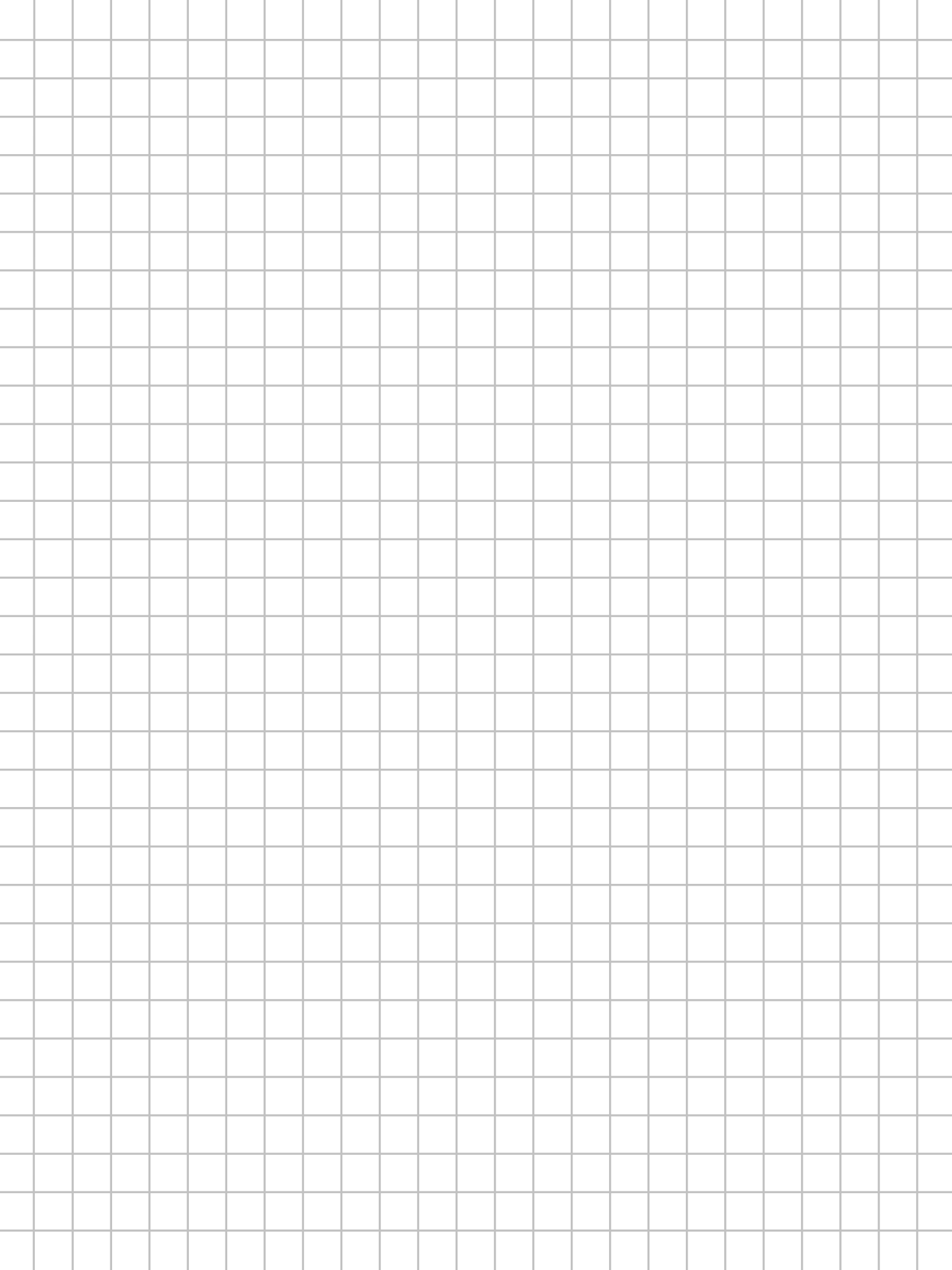
Proof

$\Leftarrow \checkmark$

$\Rightarrow f$ is a zero divisor. Choose some $g \in A[x]$, $f \cdot g = 0$, g has the least degree.
 $g = b_0 + \dots + b_m x^m$.

$$f \cdot g = b_0 a_0 + \dots + \underbrace{b_m a_m}_{=0} x^{m+n} = 0$$

then $(a_m g) \cdot f = a_m \cdot (g \cdot f) = 0$



$\sqrt{2}$

Let $f = \sum_{n=0}^{\infty} a_n x^n \in A[[x]]$.

a) f is invertible in $A[[x]] \iff a_0$ is invertible in A .

Proof.

$\Rightarrow$ f -invertible, then for some

$$g = \sum_{n=0}^{\infty} b_n x^n \quad f \cdot g = 1$$

$$fg = a_0 b_0 + \dots + \sum_{i+j=k} a_i b_j x^k + \dots = 1 \iff$$

$a_0 b_0 = 1$, other summands are 0.

a_0 is invertible in A

$\Leftarrow$ a_0 is inv. in A . Show that

$f = a_0 + \sum_{n=1}^{\infty} a_n x^n$ is inv. in $A[[x]]$

$$g = b_0 + \sum_{n=1}^{\infty} b_n x^n, \quad \underline{b_0} = a_0^{-1}.$$

$$a_0 b_1 + a_1 b_0 = 0$$

$$b_0 | a_0 b_1 = -a_1 b_0$$

$$\underline{b_1} = -b_0^{-1} a_1.$$

for b_k :

$$\sum_{i+j=k} a_i b_j = a_0 b_k + \dots + a_k b_0 = 0$$

$$b_k + b_0 a_1 b_{k-1} + \dots + b_0^2 a_k = 0$$

$$b_k = -(b_0 a_1 b_{k-1} + \dots + b_0^2 a_k) \quad \text{is}$$

a recurrent formula for g to be invertible for f .



$\sqrt{2}$

f is nilpotent $\Rightarrow a_n$ is nilpotent. $\forall n$

Proof

If f is nilpotent, then for some m $f^m = 0$.

$$f^m = a_0^m + \dots = 0 \quad \text{then coeff.}$$

of each summand is 0, $a_0^m = 0$

a_0 is a nilpotent.

2) Assume $a_0, \dots, a_k$ are nilpotents.

3) Prove a_k is a nilpotent

$a_0, \dots, a_{k-1}$ - nilp., then $a_0, a_1x, \dots, a_{k-1}x^{k-1}$ are nilps. $P = f - (a_0 + a_1x + \dots + a_{k-1}x^{k-1})$ is nilp. $P^m = 0$ for some m .

$$P = a_k x^k + a_{k+1} x^{k+1} + \dots = x^k (a_k + a_{k+1}x + \dots)$$

$$P^m = x^{km} (a_k + \dots)^m = 0$$

$a_k^m + \dots = 0$, then a_k is nilp.

Then a_n is nilp by induction.